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METHOD FOR CELL RESELECTION AND USER EQUIPMENT

Abstract

A method for cell reselection is provided. The method includes using user equipment (UE) to determine a key performance indicator (KPI) according to the scenario that the UE is in. The method further includes the UE measuring the KPI of the source cell, which is the serving cell of the UE. The method further includes the UE estimating the KPI of a target cell according to at least one parameter of the target cell. The method further includes the UE determining to camp on the target cell according to the KPI of the source cell and the KPI of the target cell.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS
[0001] This application claims the benefit of China Application No. 202410204873.8, filed Feb. 23, 2024, the entirety of which is incorporated by reference herein.
BACKGROUND OF THE INVENTION
Field of the Invention
[0002] The present invention relates to wireless communication, and, in particular, to cell reselection based on an application scenario of the user equipment.
Description of the Related Art
[0003] Cell reselection is a procedure allowing user equipment (UE) to change cells after the UE has camped on a cell. When the UE camps on a serving cell and is in an idle mode, the UE may measure the parameters and attributes of the serving cell and neighboring cells. Then, the UE may determine to reselect a target cell among the neighboring cells when the measurement results indicate that the target cell is better than the serving cell. However, the UE is used to refer the network configuration to select the cell. In other words, whether the UE can switch to another cell is restricted by the network configuration. For example, when the UE is served by a cell with weak signal strength, there could be another cell with good signal quality in the inter-frequency absolute radio-frequency number (ARFCN). However, reselection is blocked by the network configuration (e.g. due to the ARFCN priority). Furthermore, the network configuration may not be proper in every UE application scenario. An improper network configuration will prevent the UE from changing to a more suitable cell and thus degrade the performance of the UE and impact user experience.
[0004] A solution is sought to improve the cell reselection procedure.
BRIEF SUMMARY OF THE INVENTION
[0005] An embodiment of the present invention provides a method for cell reselection. The method includes using user equipment (UE) to determine a key performance indicator (KPI) according to the scenario that the UE is in. The method further includes the UE measuring the KPI of the source cell, which is the serving cell of the UE. The method further includes the UE estimating the KPI of a target cell according to at least one parameter of the target cell. The method further includes the UE determining to camp on the target cell according to the KPI of the source cell and the KPI of the target cell.
[0006] An embodiment of the present invention provides a kind of user equipment (UE). The UE includes a processor and a memory. The memory is configured to store a program. The processor is configured to drive the program to execute tasks. The tasks include determining a key performance indicator (KPI) according to a scenario that the UE is in. The tasks further include measuring the KPI of a source cell, which is a serving cell of the UE. The tasks further include estimating the KPI of a target cell according to at least one parameter of the target cell. The tasks further include determining to camp on the target cell according to the KPI of the source cell and the KPI of the target cell.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0007] The present invention can be more fully understood by reading the subsequent detailed description and examples with references made to the accompanying drawings, wherein:
[0008] FIG. **1** shows an illustrative diagram of a communication system according to the embodiments of the present disclosure;

[0009] FIG. **2** shows a block diagram of an electronic device according to the embodiments of the present disclosure;
[0010] FIG. **3** shows a flow diagram of a method according to the embodiments of the present disclosure; and
[0011] FIG. **4** shows a flow diagram of a method according to the embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE INVENTION

[0012] The following description is made for the purpose of illustrating the general principles of the invention and should not be taken in a limiting sense. The scope of the invention is best determined by reference to the appended claims.

[0013] FIG. **1** shows an illustrative diagram of a communication system **100** according to the embodiments of the present disclosure. Communication system **100** includes a user equipment (UE) **110**, a base station **120**, and a base station **130**. For example, the base station **120** and the base station **130** may be access points, access terminals, evolved Node-Bs (eNBs), or gNodeBs (gNBs). The base station **120** provides wireless communication to multiple UEs within a geographic area **121**, and the base station **130** provides wireless communication to multiple UEs within a geographic area **131**. In some embodiments, a cell is a geographic area served by a base station, such as the geographic areas **121** and **131** (also referred to as cells **121** and **131**). In some embodiments, a cell is at least one channel served by a base station, and different cells may be served by the same or different base stations. Different cells may have different channel conditions, network configurations, and/or ARFCN priority.

[0014] In one example, the UE **110** camps on the cell **121**, and the cell **121** is referred to as the serving cell or source cell of the UE **110**. Thus, the UE **110** communicates with the base station **120**. For example, the UE **110** may connect to a network (such as Internet) through the base station **120** or receive paging from the base station **120**. When the UE **110** is in an idle mode (such as an RRC_IDLE mode), the UE **110** may measure the parameters and attributes of the cell **121** and the cell **131**. The cell **131** is referred to as a target cell, a neighboring cell, or a candidate cell. Based on the measurement result, the UE **110** may determine to change/switch/handover to the cell **131** and camp on the cell **131**.

[0015] FIG. **2** shows a block diagram of an electronic device **200** according to the embodiments of the present disclosure. Electronic device **200** may perform various functions to implement processes and methods described herein, such as methods for cell reselection in accordance to embodiments of the present disclosure. Electronic device **200** may be a part of a user equipment (UE) or implemented in a user equipment (the electronic device **200** is also referred to as a UE **200** below). For example, UE **200** may be a mobile apparatus, a wearable apparatus, a wireless communication apparatus, or a computing apparatus. In some embodiments, UE **200** is implemented in a smartphone, a smartwatch, a tablet computer, a notebook computer, or a home control center. Electronic device may be implemented in the form of one or more integrated-circuit (IC) chips such as, one or more processors. The UE **200** includes a processor **210**, a memory **220**, a transceiver **230**, a KPI determination module **240**, a measurement module **250**, a KPI estimation module **260**, and a cell reselection module **270**.

[0016] The processor **210** controls operations of the UE **200**. The processor **210** provides the required process ability to perform operating systems, programs, software, modules, applications, and functions of the UE **200**. In some embodiments, processor **210** may be implemented in the form of hardware with electronic components including transistors, diodes, capacitors, resistors, or inductors. These components are configured and arranged to achieve specific purposes in accordance with the present disclosure. In other words, the processor **210** is a special-purpose machine specifically configured to perform specific tasks including method of the present disclosure. For example, the processor **210** may be a general purpose micro-processor, a central processing unit, the combination of the general purpose processor and special purpose processor, and/or related chip set.

[0017] The memory **220** stores data required by the processor **210**. The memory **220** may include non-volatile memories, such as read only memory (ROM) and flash memory. The memory **220** may also include volatile memories, such as dynamic random access memory (DRAM) and static random access memory (SRAM). In some embodiments, the memory **220** stores a program **221** (e.g. computer-readable instruction). The program **221** can be operated by the processor **210**. When the program **221** is operated by the processor **210**, the program **221** causes the processor **210** to implement methods according to the embodiments of the present disclosure.

[0018] The transceiver **230** is capable to transmit and receive data wirelessly. The transceiver **230** is coupled with one or more antennas (not shown). The transceiver **230** receives radio frequency (RF) signals from the antenna and converts RF signals to baseband signals. The transceiver **230** also converts the baseband signals to the RF signals and sends out the RF signals through the antenna.

[0019] The KPI determination module **240** is configured to determine a KPI according to the scenario that UE **200** is in. The measurement module **250** is configured to measure the KPI of the source cell, the parameters of the target cell, and the KPI of the target cell. The KPI estimation module **260** is configured to estimate the KPI of a target cell according to at least one parameter of the target cell. The cell reselection module **270** is configured to determine to camp on the source cell or the target cell according to the KPI of the source cell and the KPI of the target cell. In some embodiments, the KPI determination module **240**, the measurement module **250**, the KPI estimation module **260**, and the cell reselection module **270** are software modules implemented by the processor **210**. For example, the processor **210** is configured to drive program **221** and implement these modules in order to execute methods of the present disclosure. In other embodiments, the KPI determination module **240**, the measurement module **250**, the KPI estimation module **260**, and the cell reselection module **270** are hardware circuits which are similar to processor **210**. Thus, the KPI determination module **240**, the measurement module **250**, the KPI estimation module **260**, and the cell reselection module **270** may be referred to as the KPI determination circuit, the measurement circuit, the KPI estimation circuit, and the cell reselection circuit, respectively. For example, the KPI determination module **240**, the measurement module **250**, the KPI estimation module **260**, and the cell reselection module **270** are chips, integrated circuits, or microprocessors.

[0020] FIG. **3** shows a flow diagram of a method **300** according to the embodiments of the present disclosure. The method **300** can be performed by the UE **200**. The method **300** starts from operation **301**. In operation **301**, the KPI determination module **240** determines a KPI according to a scenario that the UE **200** is in. Specifically, the scenario includes but not limit to whether the UE is in idle or standby mode, whether the UE is transmitting data, whether the UE is executing delay-sensitive applications, etc. The KPI determination module **240** may determine the scenario according to the operation that the UE **200** is performing. For example, the KPI determination module **240** may determine the scenario according to the mode that the UE **200** is operated in (such as idle/stand-by mode or connected mode), the application/function that the UE **200** is performing, and data that the UE **200** is transmitting or receiving. In some embodiments, the KPI determination module **240** determines the scenario according to types (such as voice and video) and sizes of data that the UE **200** is transmitting or receiving.

[0021] Then, the KPI determination module **240** can determine the KPI according to the scenario. When the UE **200** is in an idle mode or a stand-by mode, the KPI determination module **240** determines that the KPI is power consumption. When the UE **200** is in an idle mode or a stand-by mode, UE stand-by time is critical to user experience. It is desirable to extend the UE stand-by time as much as possible, and the less power consumption means that the UE has longer UE stand-by time. In some embodiments, the KPI determination module **240** determines that the KPI is a paging reception successful rate when the UE **200** is in an idle mode or a stand-by mode. On the other hand, when the UE is transmitting data, the KPI determination module **240** determines that the KPI is a data rate. When the UE is transmitting data, especially large files, user experience is affected by throughput and data transferring time, and the throughput and the data transferring time correspond to the data rate. Larger data rate leads to higher throughput and shorter data transferring time. In some embodiments, the KPI determination module **240** determines the KPI is data rate when the UE **200** is transmitting a data having a size larger than a threshold. Moreover, when the UE is performing a delay-sensitive application, the KPI determination module **240** determines that the KPI is latency. For example, the delay-sensitive application is playing online game or watching live stream. When the UE is performing the delay-sensitive application, it is desirable to have smooth interaction without stuck. As a result, shorter latency leads to better user experience.

[0022] In operation **302**, the measurement module **250** measures the determined KPI of the source cell. The source cell is the serving cell of the UE **200**. The measured KPI is then stored in the memory **220**. Because the UE **200** camps on the source cell, the KPI of source cell can be measured directly. In operation **303**, the UE **200** obtains parameters of a target cell other than the source cell. These parameters are parameters which affect the determined KPI. In some embodiments, the UE **200** measures these parameters (e.g. using measurement module **250** or other circuits). In some embodiments, the UE **200** obtains these parameters from the information (such as broadcast messages) received from the base station of the target cell. In operation **304**, the KPI estimation module **260** estimates the KPI of the target cell according to the parameters of the target cell.

[0023] When the UE **200** is in an idle mode or a stand-by mode, the parameters obtained in operation **303** include signal quality, paging cycle, or both. In some embodiments, signal quality parameters include reference signal received power (RSRP), received signal strength indicator (RSSI), reference signal received quality (RSRQ), and/or signal to interference plus noise ratio (SINR). The UE needs to operate more process to ensure link reliability when the signal quality is poor. Thus, the higher the signal quality is, the lower the estimated power consumption of target cell is. Moreover, the UE can stay in sleep mode for a longer time and save more power when the paging cycle is long. Thus, the longer the paging cycle is, the lower the estimated power consumption of target cell is.

[0024] When the UE is transmitting data, the parameters obtained in operation **303** include signal quality, rank, bandwidth (BW), allocated resource block (such as the size and number of the allocated resource block), and/or allocated component carrier (CC) number (number of carrier aggregation (CA)). In some embodiments, signal quality parameters include a RSRP, RSSI, and/or RSRQ. In this disclosure, rank may refer to the number of independent data streams simultaneously transmitted/received in a multiple-input-multiple-output (MIMO) transmission. The data rate becomes larger when the signal quality is good and when there's more available resource. Thus, the estimated data rate of target cell is larger when the signal quality is higher, the rank is higher, bandwidth is larger, allocated resource block is more or larger, and/or allocated component carrier number is larger.

[0025] When the UE is performing a delay-sensitive application, the parameters obtained in operation **303** include signal quality, subcarrier spacing (SCS), and/or symbol duration. In some embodiments, signal quality parameters include a RSRP, RSSI, and/or RSRQ. The probability of re-transmission becomes lower when the signal quality is good. Thus, the higher the signal quality is, the shorter the estimated latency of target cell is. Moreover, the shorter the symbol duration is, the shorter the estimated latency of target cell is. Similarly, the larger the subcarrier spacing is, the shorter the estimated latency of target cell is.

[0026] In some embodiments, the KPI estimation module **260** may stores the correspondence relationships between parameters and the corresponding KPI of the parameters. The correspondence relationships may be obtained from statistics, simulations, or experiments. In other embodiments, the KPI estimation module **260** may use algorithms or mathematical formulas to estimate the KPI according to the parameters.

[0027] In operation **305**, the cell reselection module **270** determines whether to camp on the target cell according to the stored KPI of the source cell and the estimated KPI of the target cell. If no, the method **300** proceeds to operation **306**. If yes, the method **300** proceeds to operation **307**. The cell reselection module **270** determines whether the KPI of the source cell is better than the KPI of the target cell and decides to camp on the cell with better KPI. When the UE **200** is in an idle mode or a stand-by mode, the cell reselection module **270** determines to camp on the cell with lower power consumption. When the UE **200** is transmitting data, the cell reselection module **270** determines to camp on the cell with higher data rate. When the UE **200** is performing a delay-sensitive application, the cell reselection module **270** determines to camp on the cell with shorter latency.
[0028] In operation **306**, the UE **200** stays in the source cell. Then, the method **300** goes back to operation **301**. In operation **307**, the UE **200** handovers to the target cell. In some embodiments, the UE **200** transmits a message indicating the preference to handover to the target cell to the base station of the source cell then performs the handover procedure. Thus, the UE **200** camps on the target cell after operation **307**.
[0029] In operation **308**, the measurement module **250** measures the determined KPI of the target cell. Because the UE **200** camps on the target cell now, the KPI of the target cell can be measured directly. As mentioned above, when the UE **200** is in an idle mode or a stand-by mode, the measurement module **250** measures the power consumption of the target cell. When the UE **200** is transmitting data, the measurement module **250** measures the data rate of the target cell. When the UE **200** is performing a delay-sensitive application, the measurement module **250** measures the latency of the target cell.
[0030] In operation **309**, the cell reselection module **270** determines whether to camp on the source cell according to the stored KPI of the source cell and the measured KPI of the target cell. If yes, the method **300** proceeds to operation **310**. If no, the method **300** goes back to operation **301**. Operation **309** is similar to operation **305**. As mentioned above, the cell reselection module **270** determines whether the stored KPI of the source cell is better than the measured KPI of the target cell and decides to camp on the cell with better KPI. In operation **310**, the UE **200** handovers to the source cell. That is, if the measured KPI of the target cell is worse than the stored KPI of the source cell, the UE **200** moves back to the original cell.
[0031] FIG. **4** shows a flow diagram of a method **400** according to the embodiments of the present disclosure. The method **400** can be performed by the UE **200**. The method **400** starts from operation **401**. In operation **401**, the KPI determination module **240** determines a KPI according to a scenario that the UE **200** is in. In operation **402**, the measurement module **250** measures the KPI of the source cell. The source cell is the serving cell of the UE **200**. In operation **403**, the KPI estimation module **260** estimates the KPI of a target cell according to at least one parameter of the target cell. In operation **404**, the cell reselection module **270** determines to camp on the target cell according to the KPI of the source cell and the KPI of the target cell.
[0032] Thus, the proposed method can use different KPI as cell reselection metric under different scenarios so as to optimize user experience. Moreover, in the proposed embodiments, the UE can determine whether to leave the serving cell on its own. In other words, the UE can switch to another cell without limitation of the network configuration. As a result, the UE can find a proper cell which is most suitable to the current application scenario of the UE.
[0033] While the invention has been described by way of example and in terms of the preferred embodiments, it should be understood that the invention is not limited to the disclosed embodiments. On the contrary, it is intended to cover various modifications and similar arrangements (as would be apparent to those skilled in the art). Therefore, the scope of the appended claims should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements.

Claims

1. A method for cell reselection, comprising: determining, by a user equipment (UE), a key performance indicator (KPI) according to a scenario that the UE is in; measuring, by the UE, the KPI of a source cell, which is a serving cell of the UE; estimating, by the UE, the KPI of a target cell according to at least one parameter of the target cell; and determining, by the UE, to camp on the target cell according to the KPI of the source cell and the KPI of the target cell.
2. The method as claimed in claim 1, wherein determining the KPI according to the scenario which the UE is in comprises: determining that the KPI is power consumption when the UE is in an idle mode or a stand-by mode.
3. The method as claimed in claim 2, wherein determining to camp on the target cell according to the KPI of the source cell and the KPI of the target cell comprises: determining to camp on the target cell in response to a determination that the power consumption of the target cell is less than the power consumption of the source cell.
4. The method as claimed in claim 2, wherein the parameter comprises signal quality, paging cycle, or both.
5. The method as claimed in claim 1, wherein determining the KPI according to the scenario which the UE is in comprises: determining the KPI is data rate when the UE is transmitting data.
6. The method as claimed in claim 5, wherein determining to camp on the target cell according to the KPI of the source cell and the KPI of the target cell comprises: determining to camp on the target cell in response to the determination that the data rate of the target cell is higher than the data rate of the source cell.
7. The method as claimed in claim 5, wherein the parameter comprises signal quality, rank, bandwidth (BW), allocated resource block, and/or allocated component carrier number.
8. The method as claimed in claim 1, wherein determining the KPI according to the scenario which the UE is in comprises: determining the KPI is latency when the UE is performing a delay-sensitive application.
9. The method as claimed in claim 8, wherein determining to camp on the target cell according to the KPI of the source cell and the KPI of the target cell comprises: determining to camp on the target cell in response to the determination that the latency of the target cell is shorter than the latency of the source cell.
10. The method as claimed in claim 8, wherein the parameter comprises signal quality, subcarrier spacing (SCS), and/or symbol duration.
11. The method as claimed in claim 1, further comprising: measuring, by the UE, the KPI of the target cell after camping on the target cell; and determining, by the UE, whether to camp on the source cell according to the KPI of the source cell and the measured KPI of the target cell.
12. A user equipment (UE), comprising: a processor; and a memory, configured to store a program, wherein the processor is configured to drive the program to execute the following tasks: determining a key performance indicator (KPI) according to a scenario that the UE is in; measuring the KPI of a source cell, which is a serving cell of the UE; estimating the KPI of a target cell according to at least one parameter of the target cell; and determining to camp on the target cell according to the KPI of the source cell and the KPI of the target cell.
13. The UE as claimed in claim 12, wherein processor is further configured to: determine that the KPI is power consumption when the UE is in an idle mode or a stand-by mode; and determine to camp on the target cell in response to the determination that the power consumption of the target cell is less than the power consumption of the source cell.
14. The UE as claimed in claim 13, wherein the parameter comprises signal quality, paging cycle, or both.
15. The UE as claimed in claim 12, wherein the processor is further configured to: determine the

KPI is data rate when the UE is transmitting data; and determine to camp on the target cell in response to the determination that the data rate of the target cell is higher than the data rate of the source cell.

16. The UE as claimed in claim 15, wherein the parameter comprises signal quality, rank, bandwidth (BW), allocated resource block, and/or allocated component carrier number.

17. The UE as claimed in claim 12, wherein the processor is further configured to: determine the KPI is latency when the UE is performing a delay-sensitive application; and determine to camp on the target cell in response to the determination that the latency of the target cell is shorter than the latency of the source cell.

18. The UE as claimed in claim 17, wherein the parameter comprises signal quality, subcarrier spacing (SCS), and/or symbol duration.

19. The UE as claimed in claim 12, wherein the processor is further configured to: measure the KPI of the target cell after camping on the target cell; determine whether to camp on the source cell according to the KPI of the source cell and the measured KPI of the target cell.